

trostatic quadrupoles are used in radio antennae, and magnetic quadrupoles serve to focus beams of charged particles in particle accelerators.

The device is fabricated by etching four holes in a silicon tip, which is about 1 mm². When it is brought close to a surface, it acts on it like a water-jet vacuum cleaner. Two apertures (the 'plus' holes, or sources) emit microscopic jets of fluid onto the surface below, while the other two apertures (the 'minus' holes, or drains) suck these back into the device.

World's smallest three-axis gyroscope

STMicroelectronics has expanded its motion-sensor portfolio with the addition of the world's smallest high-performance three-axis analogue gyroscope.

The L3G3250A, with a resonant frequency beyond the audio bandwidth, is completely immune to audio noise, such as from speakers mounted close to the sensor, and mechanical vibrations coupled through the printed circuit board. In addition, its design with a single sensing structure for motion measurement along all three orthogonal axes further increases sensing accuracy. The combination of a high sensing resolution with immunity to audio noise and vibrations makes motion-sensing user interfaces more realistic in mobile phones, tablets, game consoles and other smart consumer devices.

World's first SuperSpeed USB 3.0 SATA3 bridge SoC

Renesas Electronics has introduced what it claims to be the world's first SuperSpeed USB 3.0 SATA3 bridge system-on-chip (SoC) that supports the USB-attached SCSI protocol (UASP), which significantly speeds up data throughput for large volumes of data.

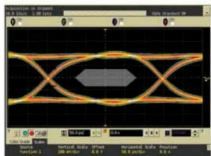
The μ PD720230 enables data transfer between a USB3.0 host system and a Serial ATA (SATA) device in external USB storage equipment.

The SoC's support for SATA Revision 3.0 enables 6Gbps high-speed data transfer. It supports USB transfer speeds of up to 5 Gbps as well as the SATA Revision 3.0 6Gbps bit rate achieving maximum data transfer performance of USB 3.0.

Lowest-capacitance transient voltage suppressors

ON Semiconductor has introduced transient-voltage-suppression devices for high-speed data lines. The ESD7008, MG2040 and ESD7104 are said to provide electrostatic discharge protection for high-speed data and video lines while offering the industry's lowest capacitance, highest signal integrity and low clamping voltages.

Designed to protect up to four high-speed differential pairs (eight lines), the ESD7008 is housed in a flow-through UDFN18 package that supports easy PCB layout and matched trace lengths necessary to



ESD7008 transient-voltage suppressor from ON Semiconductor

maintain consistent impedance for high-speed lines.

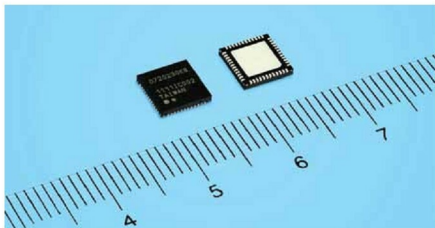
The MG2040 transient voltage suppressor is designed specifically to provide full-functionality integrated ESD protection for up to fourteen lines serving all active pins of HDMI and DisplayPort interfaces. It provides a single device replacement in applications where multiple parts would typically be used, thus saving cost and board space while improving performance.

Housed in an extremely space-efficient UDFN10 package, the ESD7104 allows easy layout and matched trace lengths between high-speed differential lines in applications such as USB 3.0, HDMI, eSATA 3.0 and DisplayPort.

World's fastest computer processor

AMD has achieved the Guinness World Record for the 'highest frequency of a computer processor,' by over-clocking the upcoming eight-core AMD FX desktop processor. The AMD FX CPU achieved a top speed of 8.429 GHz, surpassing the previous record of 8.308 GHz.

The CPU offers completely unlocked processor clock multipliers for easier over-clocking and includes exclusive AMD software through the AMD VISION Engine to fine-tune system performance. For FX branded processors, AMD has removed all the stops to enable a fully unlocked experience, giving over-clockers and PC enthusiasts complete customisation and flexibility.



SuperSpeed USB 3.0 SATA3 bridge SoC

Check efytimes.com for more news, daily